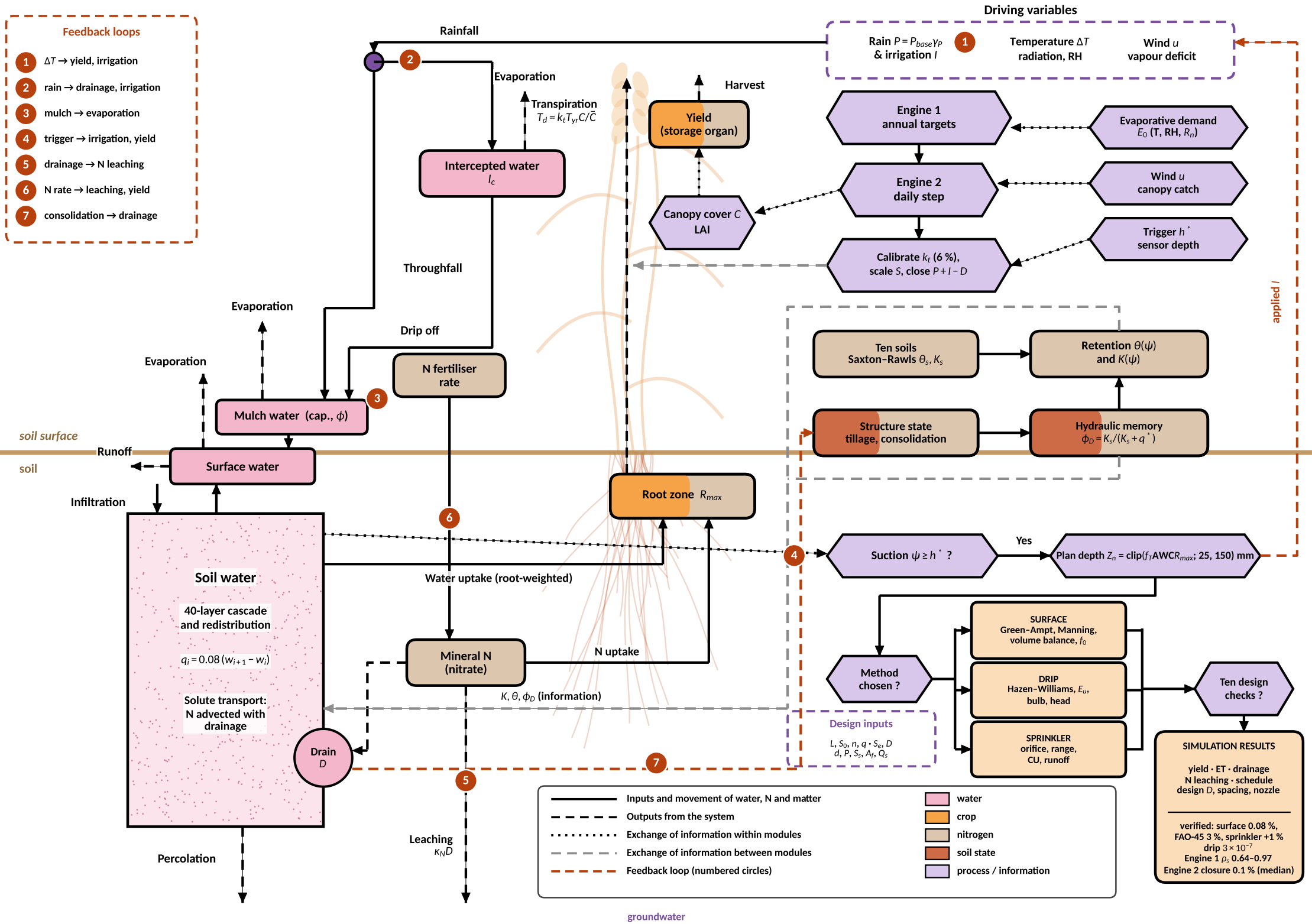

Feedback loops
1 ΔT → yield, irrigation
2 rain → drainage, irrigation
3 mulch → evaporation
4 trigger → irrigation, yield
5 drainage → N leaching
6 N rate → leaching, yield
7 consolidation → drainage
Driving variables
Rain P = P_base γ_P & irrigation I
1
Temperature ΔT radiation, RH
Wind u vapour deficit
Rainfall
2
Evaporation
Transpiration T_d = k_t T_yr C/C̄
Harvest
Yield (storage organ)
Intercepted water I_c
Canopy cover C LAI
Engine 1 annual targets
Evaporative demand E_0 (T, RH, R_n)
Engine 2 daily step
Wind u canopy catch
Trigger h* sensor depth
Calibrate k_t (6 %), scale S, close P + I − D
Throughfall
Drip off
Evaporation
Evaporation
N fertiliser rate
3
Mulch water (cap., ϕ)
soil surface
Runoff
soil
Surface water
Infiltration
Ten soils Saxton–Rawls θ_s, K_s
Retention θ(ψ) and K(ψ)
Structure state tillage, consolidation
Hydraulic memory ϕ_D = K_s/(K_s + q*)
applied I
Root zone R_max
6
Soil water
40-layer cascade and redistribution
q_i = 0.08 (w_i+1 − w_i)
Solute transport: N advected with drainage
Water uptake (root-weighted)
4
Suction ψ ≥ h* ?
Yes
Plan depth Z_n = clip(f_T AWC R_max; 25, 150) mm
Mineral N (nitrate)
N uptake
K, θ, ϕ_D (information)
Method chosen ?
SURFACE Green–Ampt, Manning, volume balance, f_0
DRIP Hazen–Williams, E_u, bulb, head
SPRINKLER orifice, range, CU, runoff
Ten design checks ?
Design inputs
L, S_0, n, q · S_e, D d, P, S_s, A_f, Q_s
Drain D
7
5
SIMULATION RESULTS
yield · ET · drainage
N leaching · schedule
design D, spacing, nozzle
verified: surface 0.08 %, FAO-45 3 %, sprinkler +1 % drip 3 × 10^−7
Engine 1 ρ_s 0.64–0.97
Engine 2 closure 0.1 % (median)
Percolation
Leaching κ_N D
Inputs and movement of water, N and matter
Outputs from the system
Exchange of information within modules
Exchange of information between modules
Feedback loop (numbered circles)
water
crop
nitrogen
soil state
process / information
groundwater